Cameron Souza

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RESEARCH SUMMARY

I study how carbon, nitrogen and phosphorus cycle through bounded biogeochemical systems, where a living community's element budget depends on exchange across a defined boundary. My dissertation develops a hierarchical Bayesian state-space framework that fuses satellite remote sensing and public monitoring data to estimate community function and boundary fluxes with explicit uncertainty, validated across the Hawaiian Archipelago. I carry the same framework into closed and bioregenerative life support systems, and more broadly into the ecology and infrastructure needed for long-duration human presence in space.

Interests: biogeochemistry and ecological stoichiometry; hierarchical Bayesian and state-space modeling; remote sensing and multi-source data fusion; island and coral reef systems; bioregenerative life support; space ecology.

EDUCATION

PhD, Environmental Sciences (Quantitative Analysis) 2026 to present
Oregon State University, College of Earth, Ocean, and Atmospheric Sciences. Advisor: Dr. Carolyn Fonyo
- Dissertation: *A Hierarchical Bayesian State-Space Framework for Boundary-Flux Modulation of Community Function: Theoretical Foundations for Bioregenerative Closed Systems, Empirically Validated on the Hawaiian Archipelago*

M.A., Data Science
Harvard University. Magna cum laude

B.S., Marine Biology and Sustainable Aquaculture
Unity College. Magna cum laude

RESEARCH AND PROFESSIONAL EXPERIENCE

Co-founder 2026 to present
Orbital Ecology Inc.

Proposal Lead, Space Futures 2026
Proposed research center on space science, engineering and infrastructure, Oregon State University
- Developing an interdisciplinary center proposal, pitched to the Huang Collaborative Innovation Complex

Founder and Principal May 2024 to present
Terra Laboratories, San Diego, CA
- Bioregenerative hardware and inference infrastructure built on bounded-system modeling

Independent Contractor Ongoing
Environmental data science and marine technical services, San Diego, CA

Project Coordinator and Data Analyst Oct 2022 to Mar 2023
Center for Marine Debris Research, Oʻahu, HI

Marine Scientist and Data Analyst (Internship) Aug 2022 to Oct 2022
Coral Restoration Foundation, Florida Keys, FL

TEACHING

Adjunct Faculty, Distance Education 2026 to present
Unity Environmental University
- MATH 301, Statistical Modeling for Life and Environmental Sciences (regression, logistic and Poisson models in R)

PRESENTATIONS AND MANUSCRIPTS

- Souza, C. *Bounded Biogeochemical Systems: A Hierarchical Bayesian Framework from the Hawaiian Archipelago to Bioregenerative Life Support.* AGU26 Annual Meeting, San Francisco, December 2026. Abstract submitted.
- Souza, C. *Hierarchical Bayesian Mass-Balance and Stoichiometric Identification for Bounded Biogeochemical Systems: A Computational Validation Across Natural and Engineered Domains.* In preparation.

SKILLS, LICENSES AND CERTIFICATIONS

Methods: Bayesian hierarchical and state-space modeling, statistical modeling, geospatial and remote sensing analysis, R, Python

Licenses: USCG 100-Ton Master · AAUS Scientific Diver · PADI Rescue Diver · First Aid and AED